

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Most sulfur in coal is present as (solid) iron pyrites (FeS_2) • Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming)	2
01.2	• Flue gases are passed through a bed of crushed calcium carbonate (limestone) • Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) • If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster	3
01.3	• The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide • The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource	2
01.4	• Fluidised bed: increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NO_x (production prevented) • Catalytic converter: platinum / palladium catalyst chemically reduces NO_x back to nitrogen and oxygen gases in the exhaust • Urea sprayed into the hot flue / exhaust gases reacts with NO_x, converting it to nitrogen, carbon dioxide and water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	
02.1	• The time taken for half of a pollutant (released into the environment) to break down / degrade	1

Question	Marking guidance	Marks
02.2	• DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) • DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level • So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level	3
02.3	<i>Any 2 from:</i> • The gas is denser than air so it settled close to the ground / did not rise and disperse • Low wind velocity (at night) meant it was not dispersed / diluted • Short half-life so it broke down before travelling far	2
02.4	• High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) • But dispersal dilutes the pollutant so the concentration at any one point may be lower	2
02.5	<i>Any 2 from:</i> • Noise / sound • Heat / thermal pollution • Ionising radiation • Light	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
03.1	• Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) • Magnetometry • Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata • Resistivity	4
03.2	• Trial drilling is the most expensive technique per site • Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites • Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves)	3
03.3	• Different minerals emit / reflect infrared at different wavelengths • Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition	2
03.4	• Results are available immediately in the field, saving time waiting for laboratory analysis	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	• Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm⁻³ at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm⁻³ at 4 and 28 h) • The night-time minimum is lower on the second night (2.5 vs 3.5)	2

Question	Marking guidance	Marks
04.2	- Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) - During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) - At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen - Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen; warm summer water holds less oxygen and decay is rapid	4
04.3	- At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate - At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation	2
04.4	Any 2 from: - Cyanobacteria release toxins harmful to livestock, pets and people - Floating algae shade submerged / rooted macrophytes which die, disrupting food webs - Algae are an unreliable food source (die off rapidly) so food webs are poor - Blocked filters / unpleasant taste and smell in water supplies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	
05.1	- Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work)	2
05.2	$N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1)	3
05.3	- Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) - Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread	2
05.4	- A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants - Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening	2
05.5	Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
06.1	- B (about 40:40:20 sand:silt:clay) - Balanced mix gives good drainage and aeration from the sand fraction - Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation	3

Question	Marking guidance	Marks
06.2	• Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) • Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand • With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too	3
06.3	• C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather • Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A	2
06.4	• Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats) • Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
07.1	• C	1
07.2	• Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms • Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas	2
07.3	• HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere • So most do not persist long enough to reach the stratosphere where UV would release their chlorine	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
08.1	• $10^3 / 6^3 = 1000 / 216$ • = 4.6 times (accept 4.63)	2
08.2	• C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs • A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release); far from grid; scenic objections • B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting • D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy • Judgement, eg C or B; turbines spaced 3-5 blade diameters apart to avoid wind shadow	5
08.3	<i>Any 2 from:</i> • Driven by wind from any direction so no motor to turn them; suitable for turbulent urban sites • Generator can be at ground level so a lighter tower • Quieter • Operate at lower wind velocities • Blades more visible so fewer bird deaths	2

Question	Marking guidance	Marks
08.4	• No gearbox: fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	
09.1	• Wheel vibration: track polishing / grinding, sound-absorbing ballast, sound-absorbing suspension • Pantograph: aerodynamic fairing • Wheel squeal: lubrication of wheels / track on corners • Braking: composite-material brakes instead of cast iron	4
09.2	• Roads pass through / near almost all residential areas and traffic is continuous through the day, unlike airports or railways which affect limited areas	1
09.3	<i>Any 3 from:</i> • Sound-absorbing road surfaces (porous asphalt / rubber-crumb asphalt) • Acoustic fences, embankments / baffle mounds, walls or dense tree planting • Traffic management for free flow (less acceleration noise); re-routing traffic away from residential areas • Double glazing / acoustic insulation of buildings • Vehicle design: better aerodynamics, engine insulation, quieter exhausts, quieter tyres	3
09.4	<i>Any 2 from:</i> • Air bubble curtains around the pile driver absorb noise • Acoustic survey to detect cetaceans first and delay work if present; increase volume gradually (soft start) so animals can leave • Drilling instead of pile-driving	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 292-294; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter • Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity • Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
10.2	• The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Question	Marking guidance	Marks
11.1	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25